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PRACTICAL 7: ROUTER CONFIGURATION AND INTER-NETWORK COMMUNICATION

Objective

Configure routers and establish communication between networks.

Equipment

- Cisco 1941 Router
- Cisco ISR4331 Router
- Cisco Switches
- PCs

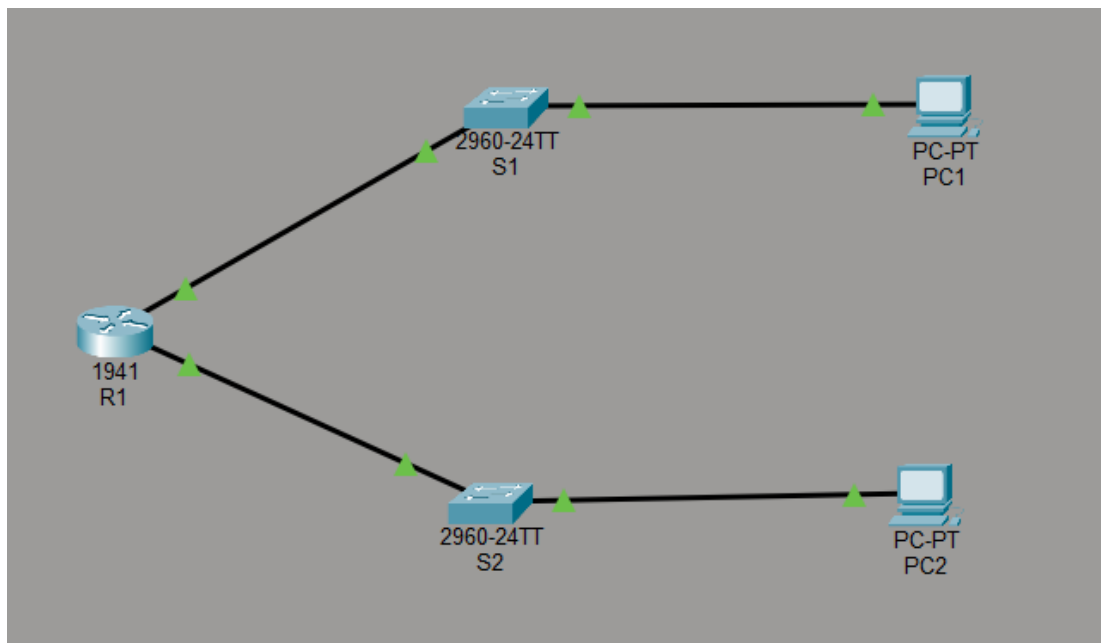


Figure 1: Full topology

```
R1
Physical Config CLI Attributes

255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip address 10.10.10.1 255.255.255.0
^
% Invalid input detected at '^' marker.

Router(config)#interface g0/0
Router(config-if)#ip address 10.10.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#interface g0/1
^
% Invalid input detected at '^' marker.

Router#enable
Router#configure terminal
^
% Invalid input detected at '^' marker.

Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/1
Router(config-if)#ip address 10.10.20.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router(config-if)#
```

Figure 2: Router 1 Configuration

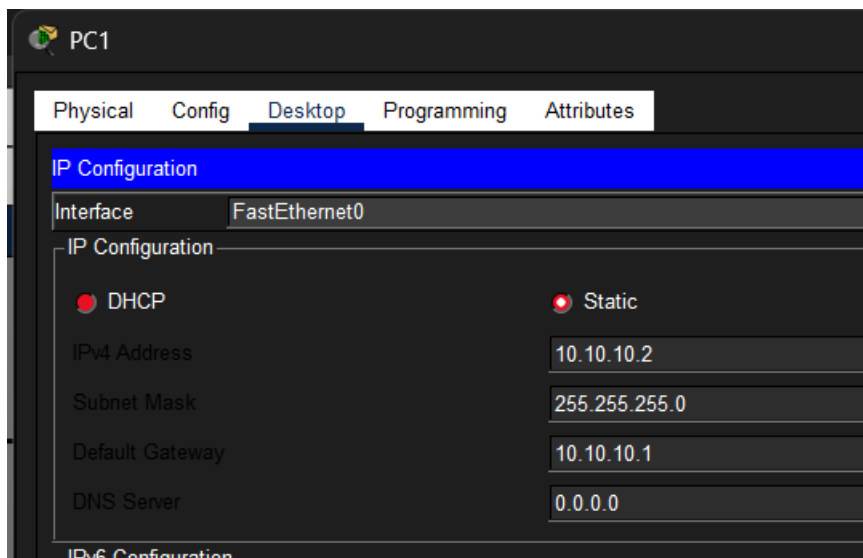


Figure 3: PC1 IP CONFIG

```
Router#show ip interface brief
Interface      IP-Address      OK? Method Status  Protocol
GigabitEthernet0/0  10.10.10.1      YES manual up      up
GigabitEthernet0/1  10.10.20.1      YES manual up      up
Vlan1           unassigned      YES unset  administratively down down
Router#
```

Figure 4: Router show ip interface brief

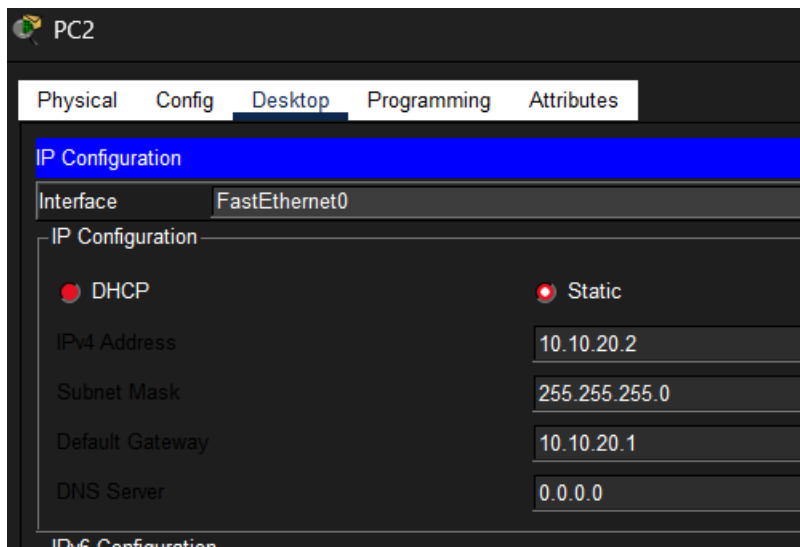
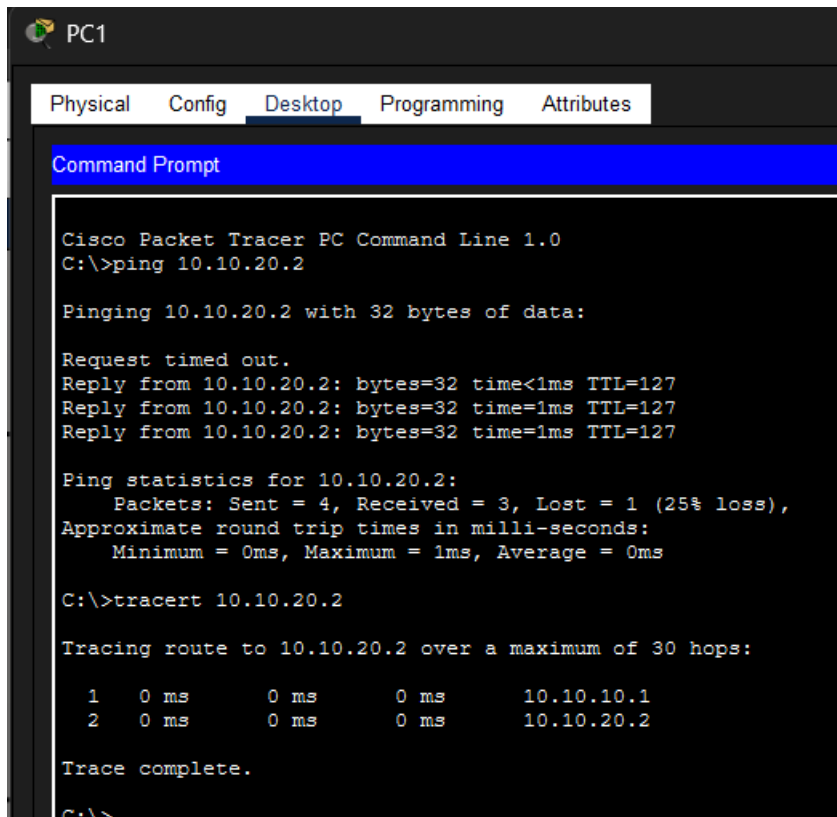


Figure 5: PC2 IP CONFIG



```
PC1
Physical  Config  Desktop  Programming  Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.10.20.2

Pinging 10.10.20.2 with 32 bytes of data:

Request timed out.
Reply from 10.10.20.2: bytes=32 time<1ms TTL=127
Reply from 10.10.20.2: bytes=32 time=1ms TTL=127
Reply from 10.10.20.2: bytes=32 time=1ms TTL=127

Ping statistics for 10.10.20.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>tracert 10.10.20.2

Tracing route to 10.10.20.2 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    10.10.10.1
  2  0 ms    0 ms    0 ms    10.10.20.2

Trace complete.

C:\>
```

Figure 6: Verification Ping Tracert

Activities

Create two networks:

Network A 1 192.168.10.0/24

Network B 1 192.168.20.0/24

Example Configuration 1 interface g0/0

2 ip address 192.168.10.1 255.255.255.0

3 no shutdown

5 interface g0/1

6 ip address 192.168.20.1 255.255.255.0

7 no shutdown

Verification Commands

1 show ip interface brief

2 ping

3 tracert

Deliverable

Successful communication between both networks.

Assessment Rubric (10 Marks) Criteria	Marks
Router configuratio n	4
Interface addressing	2
Connectivity testing	2
Documentati on	2
Total	10